

16.5 Answers

① a) $-y \vec{i} - z \vec{j} - x \vec{k}$; $x+y+z$

b) $\vec{0}$; 1

c) $\frac{y}{z^2} \vec{i} - \frac{x}{z^2} \vec{j}$

② a) $\operatorname{div} \vec{F} < 0$

b) $\operatorname{curl} \vec{F} = \vec{0}$

③ a) $f(x,y,z) = x^2y + y^2z + K$

b) $\vec{F}$ is not conservative

c) $f(x,y,z) = ye^{xz} + K$

④ Hint: Show $\operatorname{curl} \vec{F} = \vec{0}$